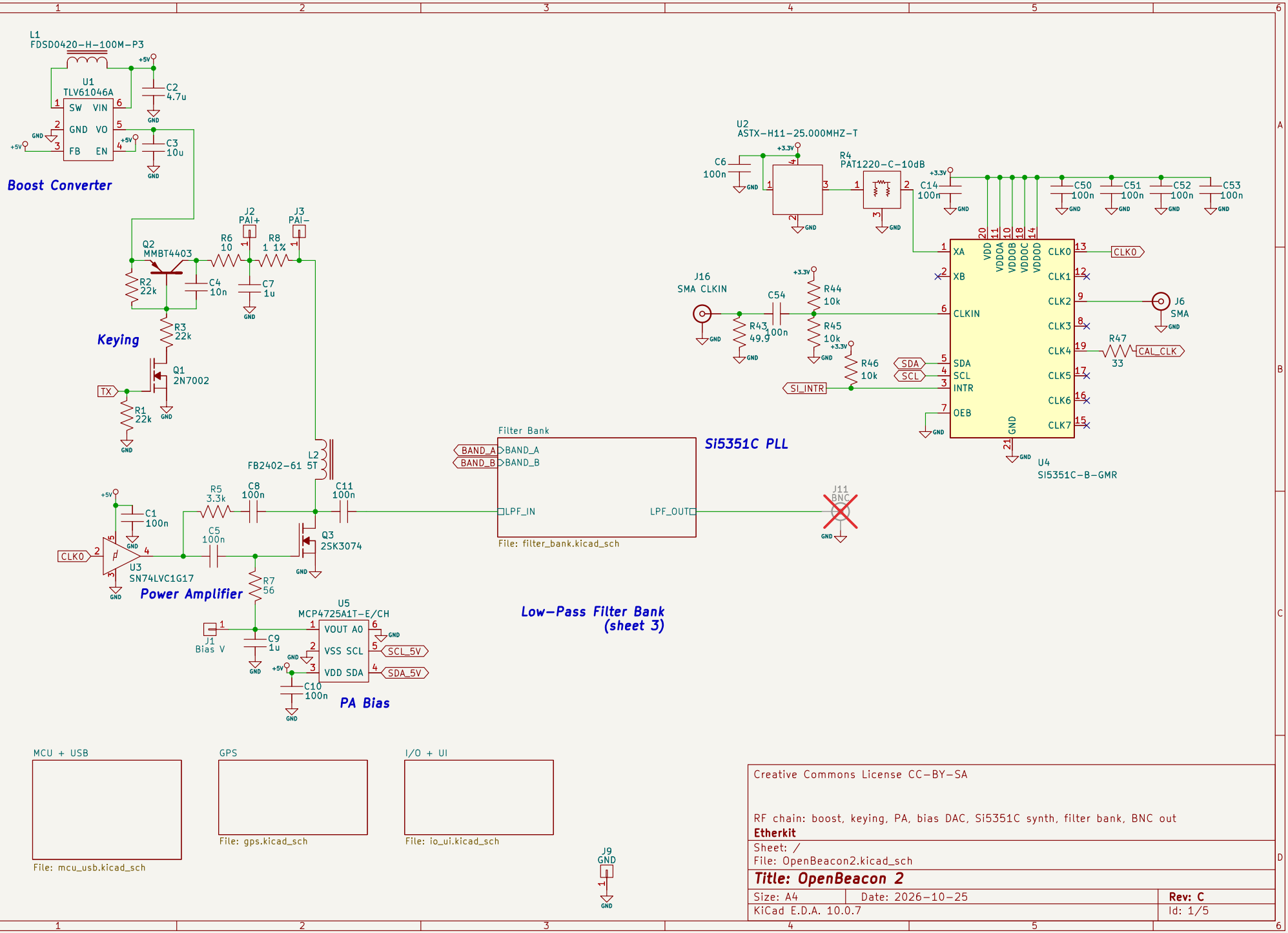
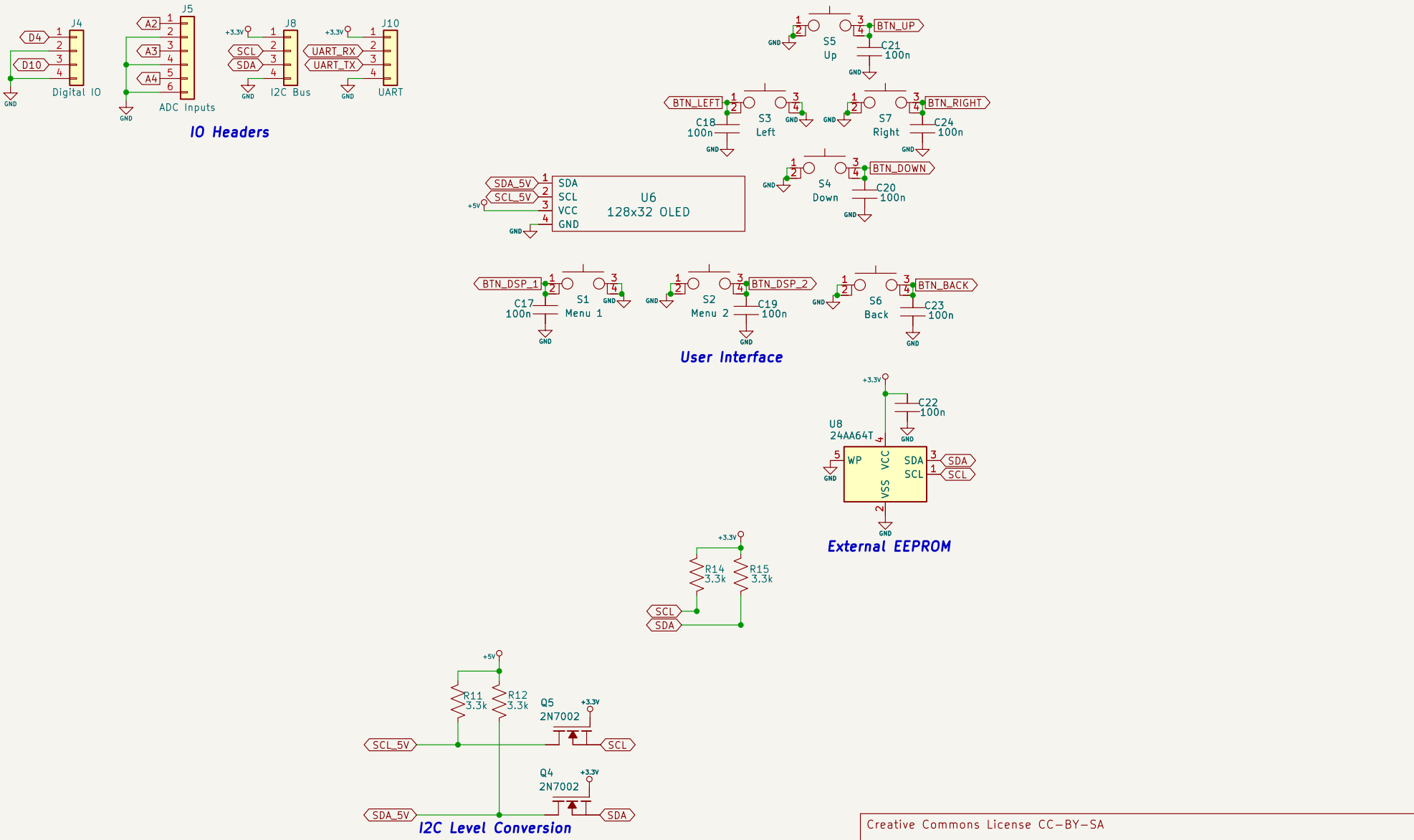

L1
FDSD0420-H-100M-P3
U1
TLV61046A
SW VIN
GND VO
FB EN
C2
4.7u
C3
10u
Boost Converter
Q2
MMBT4403
R2
22k
R3
22k
Keying
Q1
2N7002
TX
R1
22k
R6
10
R8
1 1%
C4
10n
C7
1u
J2
PAI+
J3
PAI-
L2
FB2402-61 5T
C1
100n
R5
3.3k
C8
100n
C11
100n
C5
100n
Q3
2SK3074
CLK0
U3
SN74LVC1G17
Power Amplifier
R7
56
U5
MCP4725A1T-E/CH
VOUT A0
VSS SCL
VDD SDA
SCL_5V
SDA_5V
J1
Bias V
C9
1u
C10
100n
PA Bias
Filter Bank
BAND_A
BAND_B
LPF_IN
LPF_OUT
File: filter_bank.kicad_sch
Low-Pass Filter Bank
(sheet 3)
J11
BNC
U2
ASTX-H11-25.000MHZ-T
C6
100n
R4
PAT1220-C-10dB
C14
100n
C50
100n
C51
100n
C52
100n
C53
100n
J16
SMA CLKIN
C54
100n
R43
49.9
R44
10k
R45
10k
R46
10k
SI_INTR
SDA
SCL
INTR
XA
XB
CLKIN
OEB
VDD
VDDA
VDDOB
VDDOC
VDDO
CLK0
CLK1
CLK2
CLK3
CLK4
CLK5
CLK6
CLK7
J6
SMA
R47
33
CAL_CLK
U4
SI5351C-B-GMR
Si5351C PLL
MCU + USB
File: mcu_usb.kicad_sch
GPS
File: gps.kicad_sch
I/O + UI
File: io_ui.kicad_sch
J9
GND
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RF chain: boost, keying, PA, bias DAC, Si5351C synth, filter bank, BNC out
Etherkit
Sheet: /
File: OpenBeacon2.kicad_sch
Title: OpenBeacon 2
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 1/5



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I/O headers, OLED + buttons, I2C 3.3 V / 5 V level shift, 24AA64 EEPROM

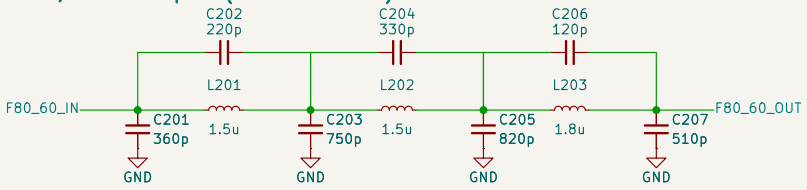
Etherkit

Sheet: /I/O + UI/
File: io_ui.kicad_sch

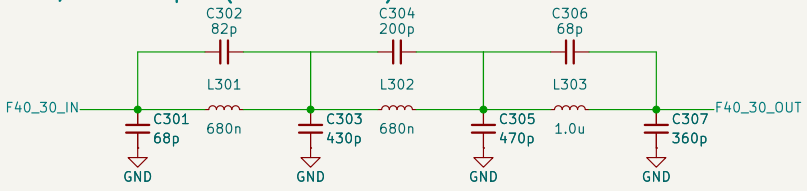
Title: OpenBeacon 2 – I/O and User Interface

Size: A4	Date: 2026-10-25	**Rev: C**
KiCad E.D.A. 10.0.7		Id: 2/5

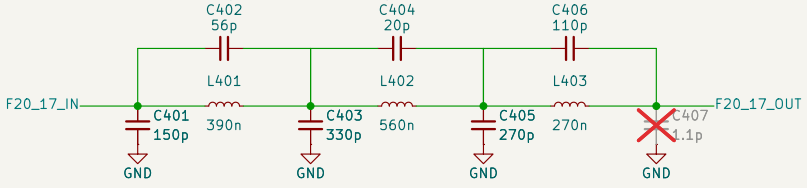
F2 80/60 m low-pass (7th-order Cauer)



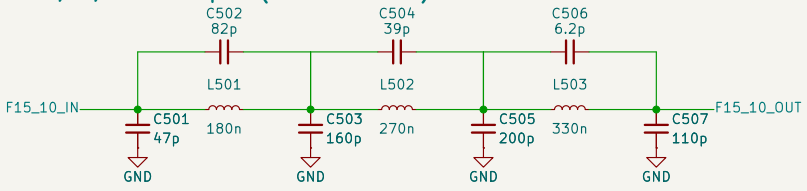
F3 40/30 m low-pass (7th-order Cauer)



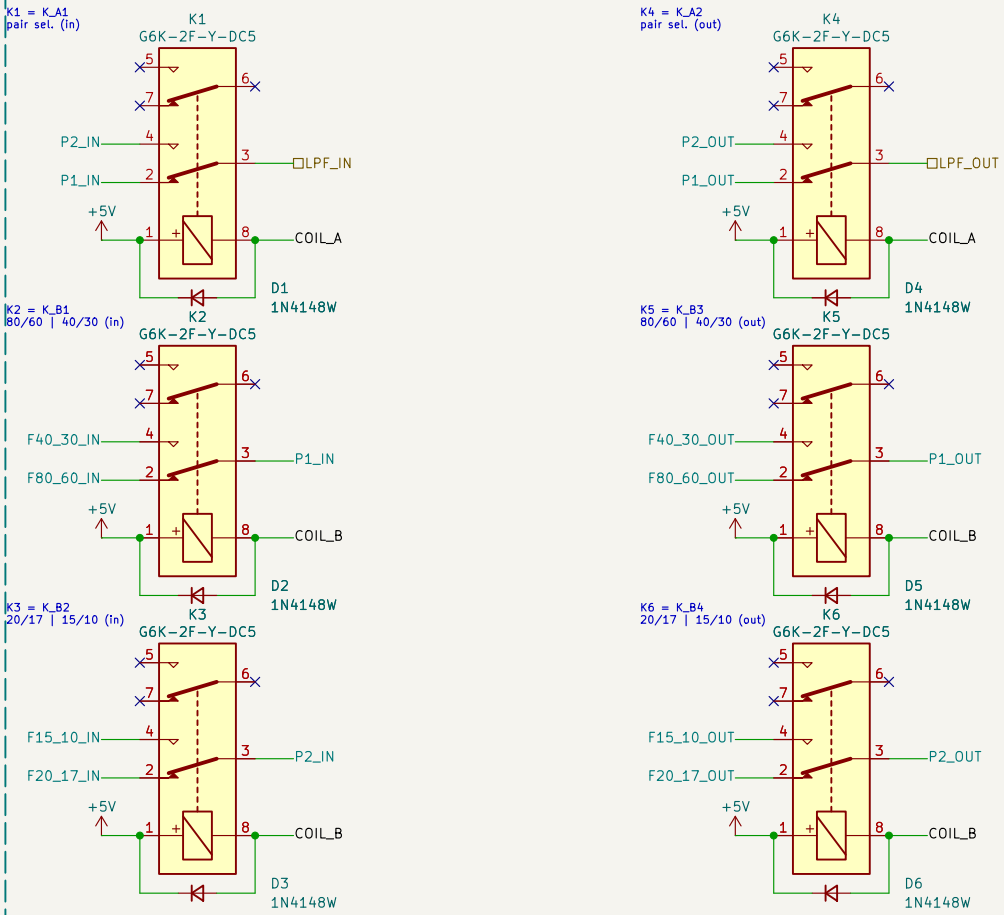
F4 20/17 m low-pass (7th-order Cauer)



F5 15/12/10 m low-pass (7th-order Cauer)



Relay tree (2 levels, input + output mirrored; all off = 80/60 m)



Relay drivers



BAND_A -> K1/K4, BAND_B -> K2/K3/K5/K6.
Q6 (BAND_A): 2 coils, 42 mA nom (5 V / 237 R), <= 49 mA at 5.25 V, -10 % coil R.
Q7 (BAND_B): 4 coils, 84 mA nom, <= 98 mA.
AO3400A: 5.7 A cont., 48 mR @ Vgs 2.5 V -> <5 mV drop.
One 1N4148W flyback per coil, at each relay.

Band code + notes

Band code (BAND_A, BAND_B):
0 0 80/60 m (all relays off, NC path, default)
0 1 40/30 m (K2, K3, K5, K6 on)
1 0 20/17 m (K1, K4 on)
1 1 15/12/10 m (all six on)
Unselected filters float (open at both ends).
+5V relay load: 0 / 84 / 42 / 127 mA nominal (<=147 mA).
Relay pins: COM right, NC lower left, NO upper left; pole 2 unused.
Input and output paths never share a relay body.
Layout: keep K1-K2-K3 and K4-K5-K6 traces short, 50 ohm CPWG over solid GND, filters next to their relays.

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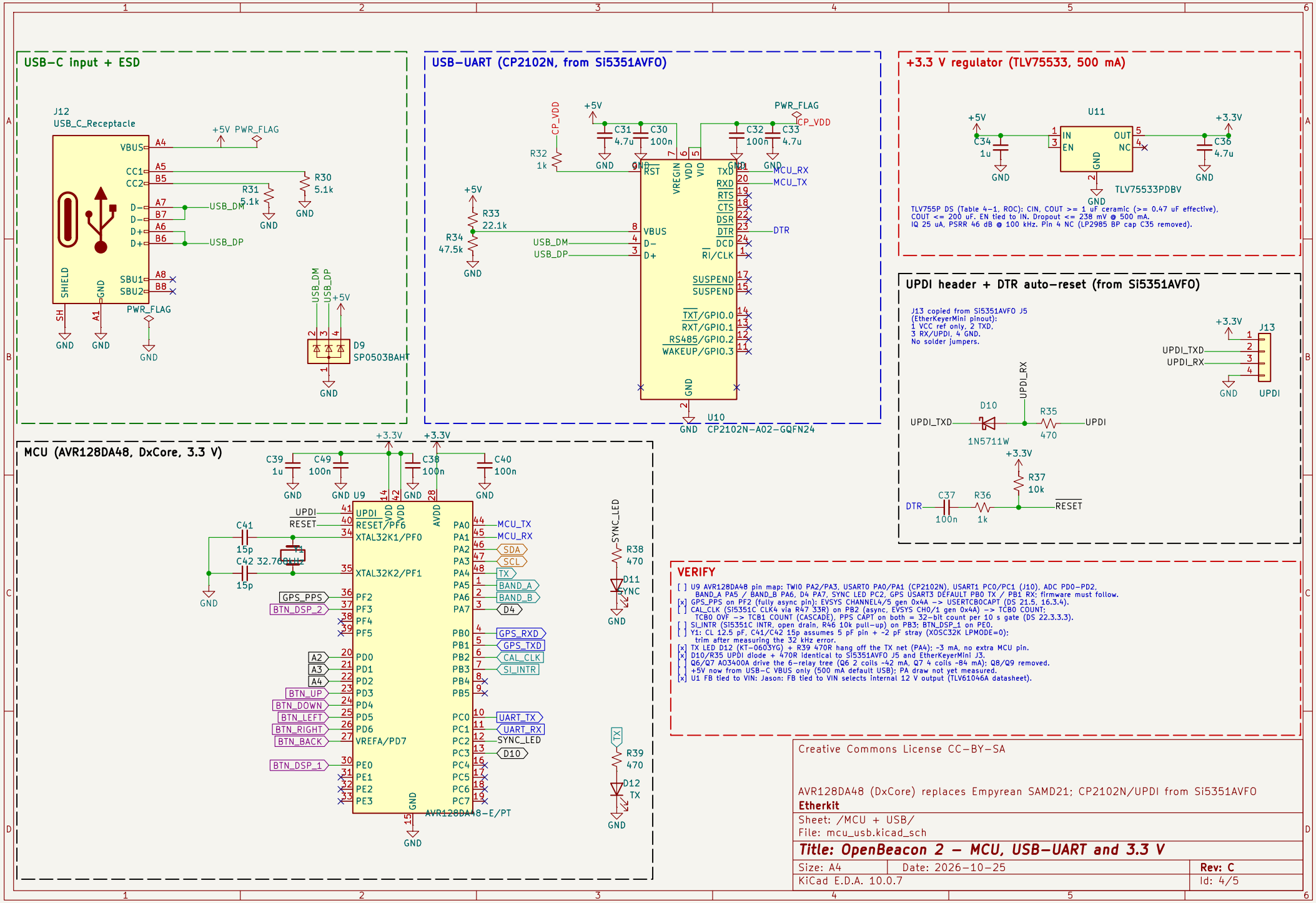
4 7th-order Cauer LPFs, 6-relay 2-level tree (2-bit BAND_A/BAND_B); unselected filters open

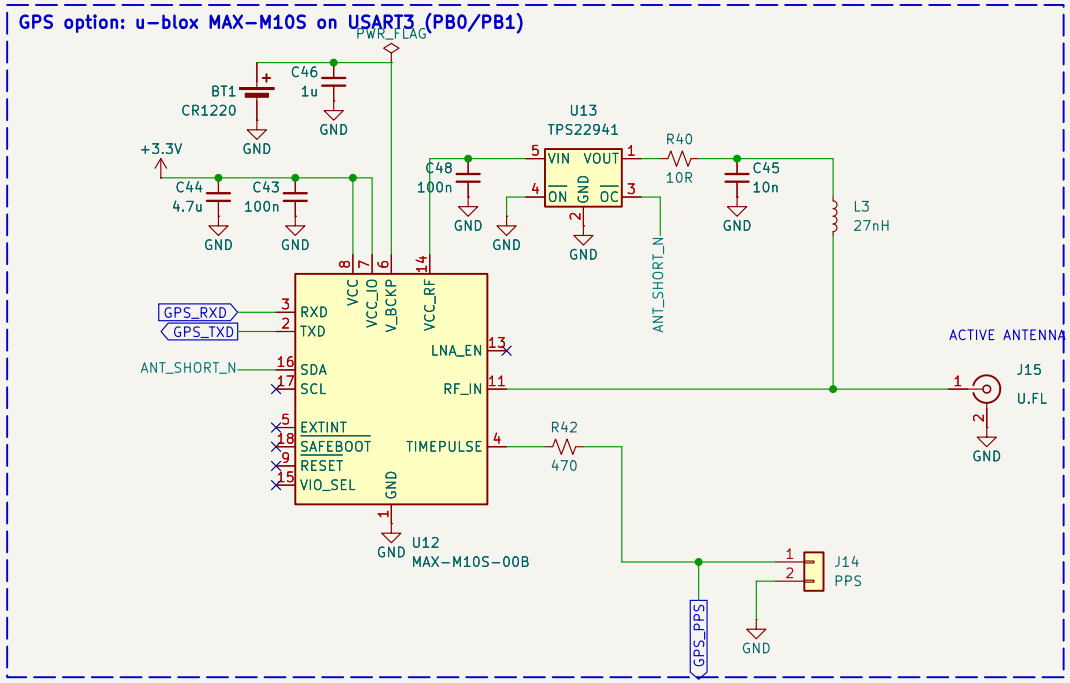
Etherkit

Sheet: /Filter Bank/
File: filter_bank.kicad_sch

Title: OpenBeacon 2 – Low-Pass Filter Bank

Size: A4	Date: 2026-10-25	Rev: C
KiCad E.D.A. 10.0.7		Id: 3/5


USB-C input + ESD
J12
USB_C_Receptacle
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
A4
A5
B5
A7
B7
A6
B6
A8
B8
SH
A1
+5V PWR_FLAG
R30
5.1k
R31
5.1k
USB_DM
USB_DP
PWR_FLAG
GND
USB_DM
USB_DP
+5V
D9
SP0503BAHT
USB-UART (CP2102N, from Si5351AVFO)
CP_VDD
+5V
PWR_FLAG
C31
4.7u
C30
100n
C32
100n
C33
4.7u
R32
1k
RST
VREGIN
VDD
VIO
GND
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
WAKEUP/GPIO.3
MCU_RX
MCU_TX
DTR
R33
22.1k
R34
47.5k
VBUS
D-
D+
U10
CP2102N-A02-GQFN24
+3.3 V regulator (TLV75533, 500 mA)
U11
C34
1u
C36
4.7u
IN
EN
OUT
NC
TLV75533PDBV
TLV755P DS (Table 4-1, ROC): CIN, COUT >= 1 uF ceramic (>= 0.47 uF effective),
COUT <= 200 uF. EN tied to IN. Dropout <= 238 mV @ 500 mA.
IQ 25 uA, PSRR 46 dB @ 100 kHz. Pin 4 NC (LP2985 BP cap C35 removed).
UPDI header + DTR auto-reset (from Si5351AVFO)
J13 copied from Si5351AVFO J5
(EtherKeyerMini pinout):
1 VCC ref only, 2 TXD,
3 RX/UPDI, 4 GND.
No solder jumpers.
J13
UPDI_TXD
UPDI_RX
UPDI
D10
1N5711W
R35
470
R37
10k
C37
100n
R36
1k
DTR
RESET
MCU (AVR128DA48, DxCore, 3.3 V)
C39
1u
C49
100n
C38
100n
C40
100n
U9
UPDI
RESET/PF6
XTAL32K1/PF0
XTAL32K2/PF1
PF2
PF3
PF4
PF5
PD0
PD1
PD2
PD3
PD4
PD5
PD6
VREFA/PD7
PE0
PE1
PE2
PE3
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB3
PB4
PB5
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
AVR128DA48-E/PT
C41
15p
C42
15p
GPS_PPS
BTN_DSP_2
A2
A3
A4
BTN_UP
BTN_DOWN
BTN_LEFT
BTN_RIGHT
BTN_BACK
BTN_DSP_1
MCU_TX
MCU_RX
SDA
SCL
TX
BAND_A
BAND_B
D4
GPS_RXD
GPS_TXD
CAL_CLK
SI_INTR
UART_TX
UART_RX
SYNC_LED
D10
SYNC_LED
R38
470
D11
SYNC
TX
R39
470
D12
TX
VERIFY
[ ] U9 AVR128DA48 pin map: TWI0 PA2/PA3, USART0 PA0/PA1 (CP2102N), USART1 PC0/PC1 (J10), ADC PD0-PD2,
BAND_A PA5 / BAND_B PA6, D4 PA7, SYNC LED PC2, GPS USART3 DEFAULT PB0 TX / PB1 RX; firmware must follow.
[x] GPS_PPS on PF2 (fully async pin); EVSYS CHANNEL4/5 gen 0x4A -> USERTCBOCAPT (DS 21.5, 16.3.4).
[ ] CAL_CLK (SI5351C CLK4 via R47 33R) on PB2 (async, EVSYS CH0/1 gen 0x4A) -> TCB0 COUNT;
TCB0 OVF -> TCB1 COUNT (CASCADE), PPS CAPT on both = 32-bit count per 10 s gate (DS 22.3.3.3).
[ ] SI_INTR (SI5351C INTR, open drain, R46 10k pull-up) on PB3; BTN_DSP_1 on PE0.
[ ] Y1: CL 12.5 pF, C41/C42 15p assumes 5 pF pin + ~2 pF stray (XOSC32K LPMODE=0);
trim after measuring the 32 kHz error.
[x] TX LED D12 (KT-0603YG) + R39 470R hang off the TX net (PA4); ~3 mA, no extra MCU pin.
[x] D10/R35 UPDI diode + 470R identical to Si5351AVFO J5 and EtherKeyerMini J3.
[ ] Q6/Q7 AO3400A drive the 6-relay tree (Q6 2 coils ~42 mA, Q7 4 coils ~84 mA); Q8/Q9 removed.
[ ] +5V now from USB-C VBUS only (500 mA default USB); PA draw not yet measured.
[x] U1 FB tied to VIN: Jason: FB tied to VIN selects internal 12 V output (TLV61046A datasheet).
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AVR128DA48 (DxCore) replaces Empyrean SAMD21; CP2102N/UPDI from Si5351AVFO
Etherkit
Sheet: /MCU + USB/
File: mcu_usb.kicad_sch
Title: OpenBeacon 2 - MCU, USB-UART and 3.3 V
Size: A4
Date: 2026-10-25
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Id: 4/5



VERIFY

- [x] Antenna short: TPS22941 limit 40–80 mA, auto-restart, avg <= 27 mA (was 320 mA).
- [x] V_BCKP: CR1220 in KH-BS1220-2-SMT (C5365932): C47/D13/R41 removed.
- [ ] BT1 footprint: draw from KH-BS1220-2-SMT DS.
- [ ] U13 TPS22941 JLC stock thin (26 pcs).
- [ ] Short peak 80 mA > VCC_RF 50 mA op. (DS T13) for <= 20 ms bursts; confirm on bench.
- [ ] V_IO ramp 25–35000 us/V (DS 4.1): measure.
- [ ] GPS_PPS MCU pin: see mcu_usb.

RECEIVER: VCC + V_IO tied to +3.3V; VIO_SEL open = 3.3 V I/O (IM 4.1.2).
C43/C44 beside pins 7/8, short GND vias; no series R > 0.2 ohm in VCC (IM 4.1.1).
RESET_N, EXTINT, SCL, SAFEBOOT_N, LNA_EN left open (DS Table 10).
SDA = ANT_SHORT_N (IM 3.3.1.2): CFG-HW-ANT_CFG_SHORTDET=1, ANT_SUP_SHORT_PIN=SDA PIO.
PPS: TIMEPULSE = SAFEBOOT_N via 1k inside (DS fn 15); nothing may hold it low at power-up, so no pull-down. R42 limits a shorted J14 to ~7 mA (I_PIO 10 mA).

ANTENNA: RF_IN is DC-blocked inside the module (IM Table 1), no external DC block.
Bias-T VCC_RF -> U13 TPS22941 -> R40 10R -> C45 10n -> L3 27 nH onto RF line (IM 3.3.1.2, 4.3.4).
U13: I_LIM 40–80 mA, 5–20 ms blanking, 40–160 ms restart (TPS2294x DS 7.5): short avg <= 27 mA, typ 7 mA (VCC_RF abs max 250 mA, DS Table 12). 25 mA antenna drop ~0.27 V -> ~2.9 V at J15.
Layout: 50 ohm CPWG pin 11 -> J15, L3 pad on the line, >= 5 mm from digital.
BACKUP: BT1 CR1220 (3 V, 35 mAh, user-fitted, NOT rechargeable, no charger) on V_BCKP; C46 1u at pin 6 (IM 4.1.3). 35 mAh / 28 uA (DS Table 17) = ~1250 h (~52 days) powered off.

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MAX-M10S on USART3 (PB0/PB1), PPS to J14, current-limited active-antenna bias-T, CR1220 V_BCK

Etherkit

Sheet: /GPS/
File: gps.kicad_sch

Title: OpenBeacon 2 – GPS (u-blox MAX-M10S)

Size: A4	Date: 2026-10-25	**Rev: C**
KiCad E.D.A. 10.0.7		Id: 5/5